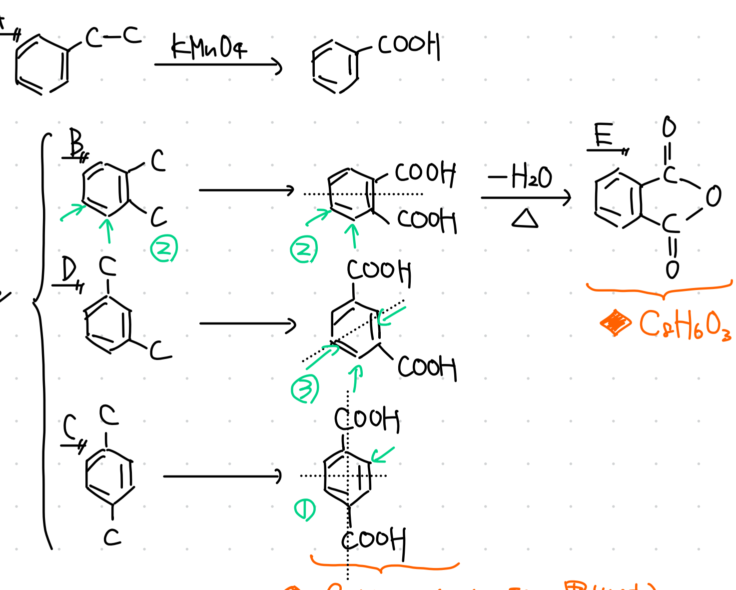
4-[1]　C_8H_{10}　$U=4$  ×1

エチルベンゼン

o, m, p-キシレン



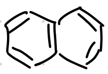
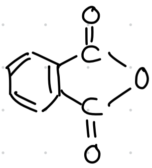
$C_8H_6O_3$（無水フタル酸）

$C_8H_6O_4$（フタル酸の異性体）

問2　(Ox)　$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$　) ×9

(Red)　$C_{10}H_8 + 7H_2O \longrightarrow C_8H_4O_3 + 2CO_2 + 18H^+ + 18e^-$　) ×2

---

2  $+ 9O_2 \longrightarrow$ 2  $+ 4CO_2 + 4H_2O$ //

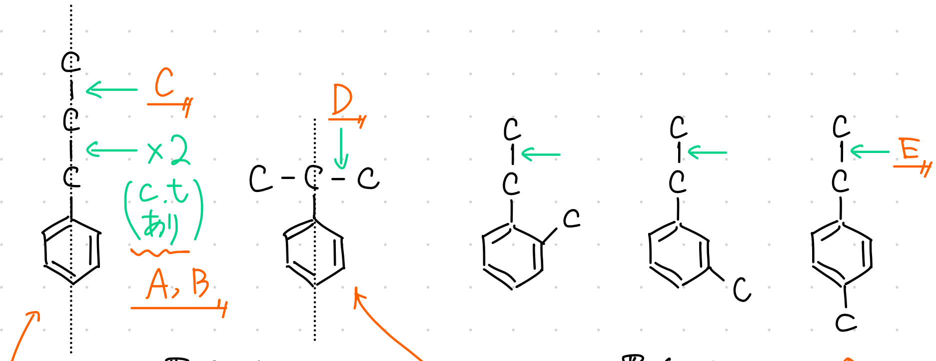
（酸化だから、余ったC $\xrightarrow{O_2}$ CO_2,
H $\xrightarrow{O_2}$ H_2O になると考えて、
係数合わせても良い）

4-図　C_9H_{10}　$U=5$

 ×1 + C=C×1

(U=4)　(U×1)

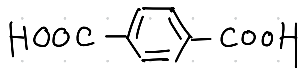
問2.



<1置換体>　　　　<2置換体>

A, B, C $\xrightarrow{H_2}$ F

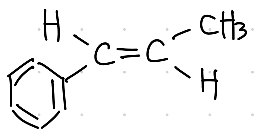
D $\xrightarrow{H_2}$ G : 7×ン

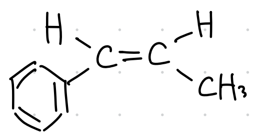
E $\xrightarrow{H_2}$ H $\xrightarrow[O_2]{Ox}$ 

A, B : シス・トランス異性体

融点 A > B

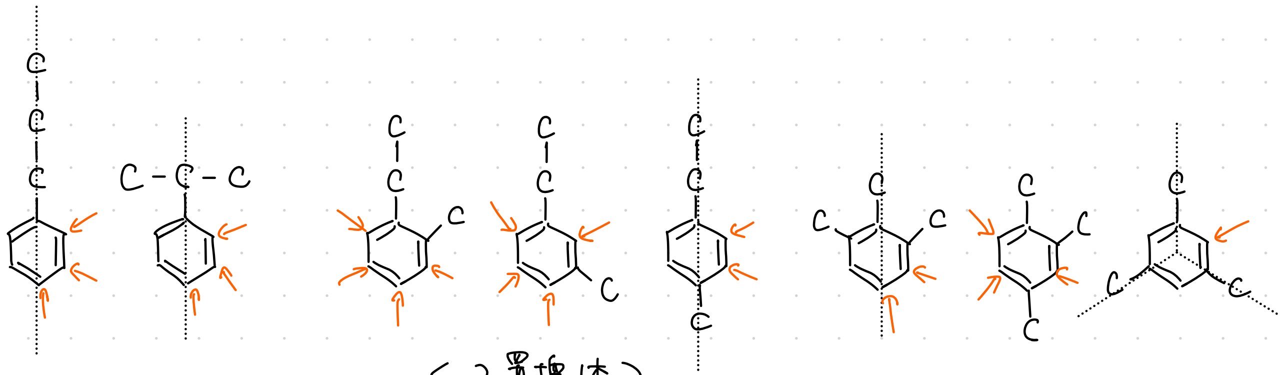
トランス　シス

A : 

B : 

問1. 8

問3. 22



<1置換体>　Cl置換 : 6

<2置換体>　Cl置換 : 10

<3置換体>　Cl置換 : 6

4-③

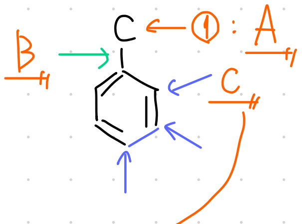
問1　$C_lH_mO_n + \underline{\quad\quad} O_2 \longrightarrow l\,CO_2 + \frac{m}{2}H_2O$

条件より　$l : \frac{m}{2} = 7 : 4$　∴ $(l, m) = (7, 8)$

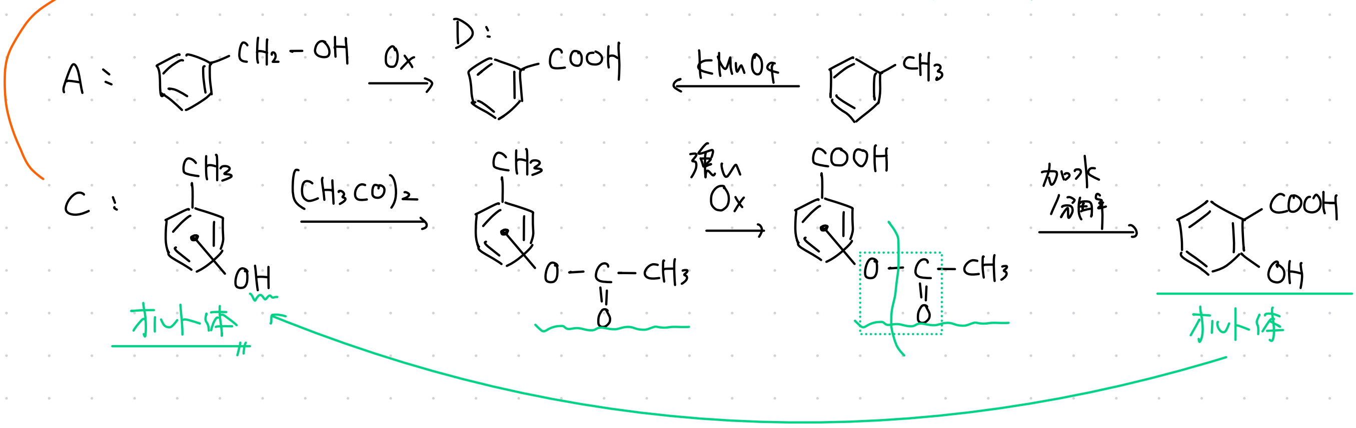
∴ $(C_7H_8O_n)_k = (92 + 16n)k = 108$

∴ $(n, k) = (1, 1)$　　　$\underline{C_7H_8O}$

問2.　C_7H_8O　$U = 4$ ⌬×1 ｛ アルコール / フェノール類 / エーテル

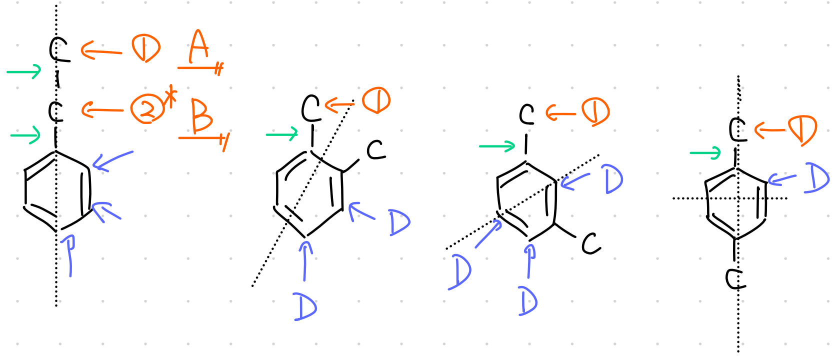


	A	B	C
		エーテル	
Na	H_2↑	×	
NaOH	×		とける
	アルコール		フェノール類



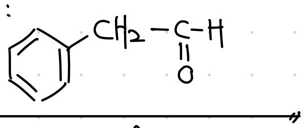
4-$\boxed{4}$　$C_8H_{10}O$　U=4  ×1

- アルコール
- フェノール類
- エーテル

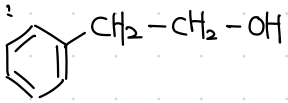


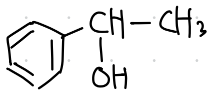
第1級アルコール

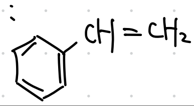
	A	B	C	D
置換基	1, 2			3
おだやか Ox	E アルデヒド			
脱水	F	F	A とBは同一C骨格	
$FeCl_3$ aq			−	+

E: 

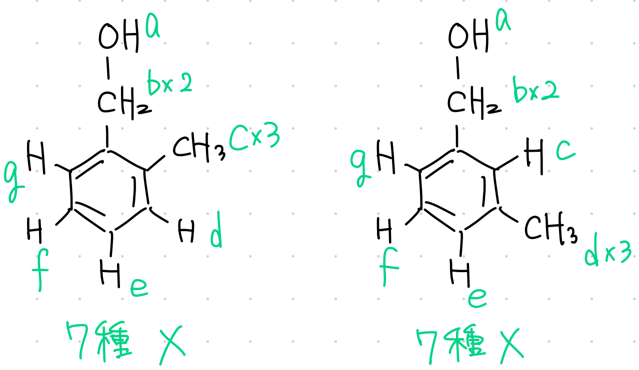
↑ おだやか酸化

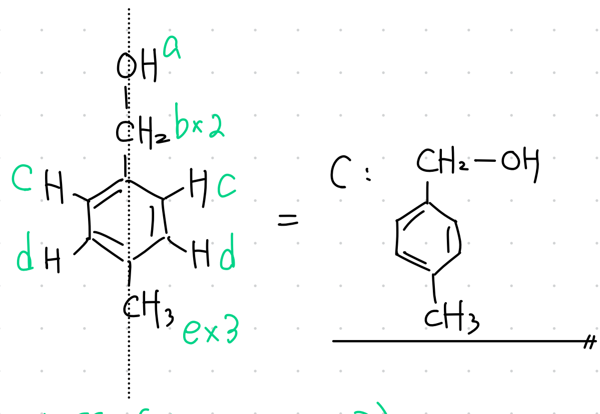
A: 

B: 

→ F: 

C : 1or 2置換体のアルコール

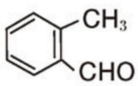
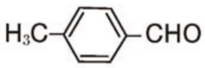


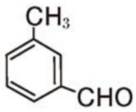
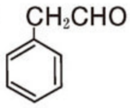
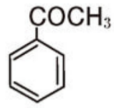


問2

D: フェノール類 (3置換体), 6

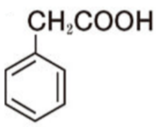
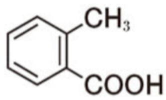
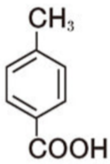
4 - 5

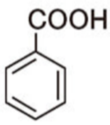
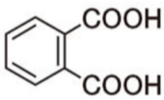
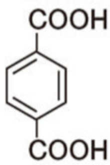
(1) A  B 

C  D  E 

(2) **G は *p*-置換体であるため，*o*-，*m*-置換体よりも分子の形が対称的である。したがって，結晶化しやすいため，融点は最も高くなる。**

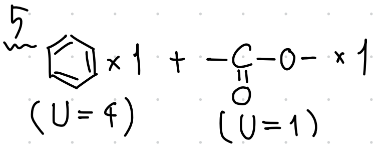
4 - 6

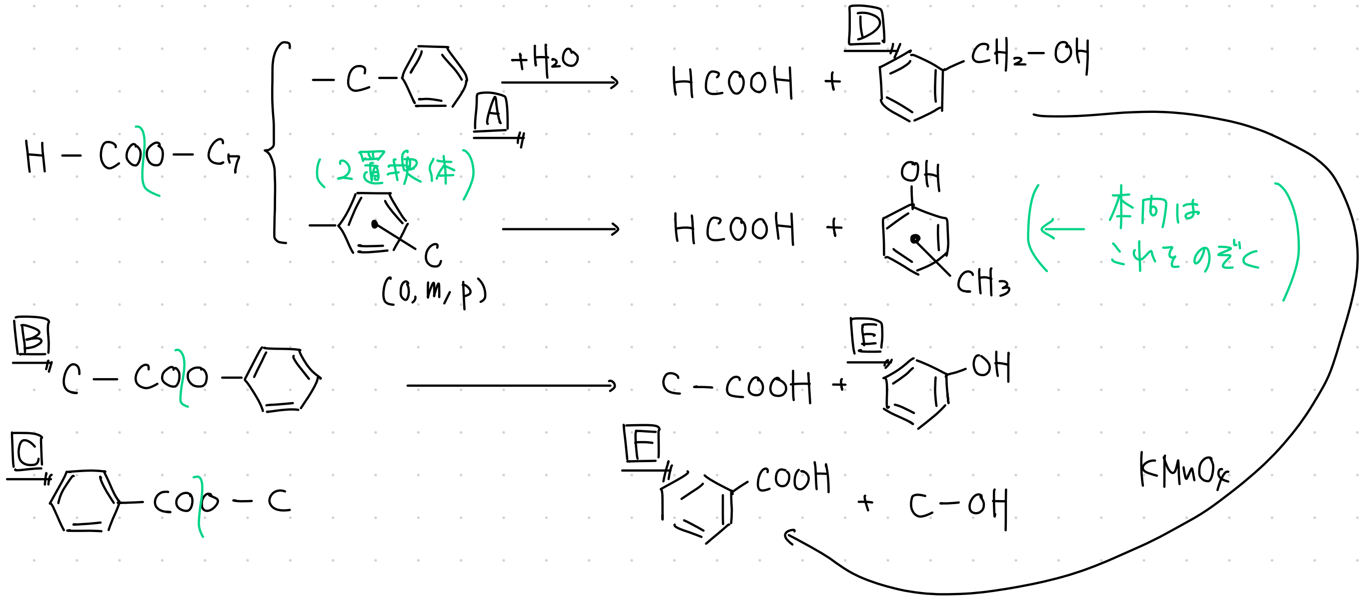
A  B  C 

D  E  F 

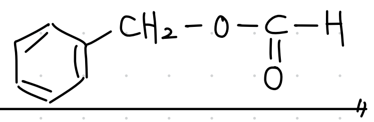
4-$\boxed{7}$

問1 $C_8H_8O_2$ $U=5$

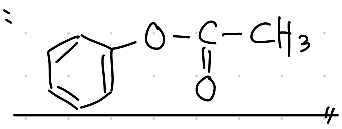




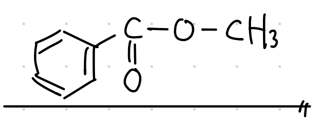
A:



B:

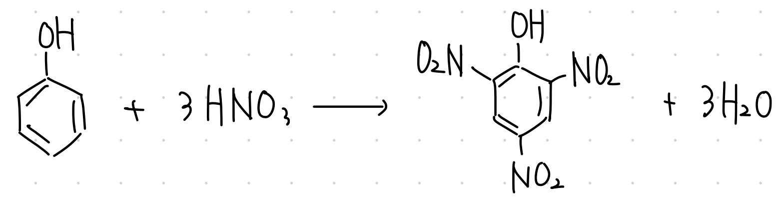


C:



問2. E

問3



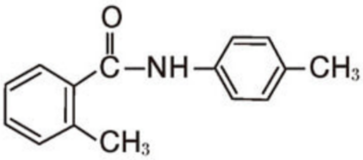
ピクリン酸

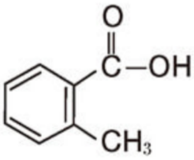
(2,4,6-トリニトロフェノール)

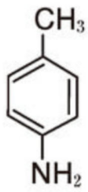
4 – $\boxed{8}$

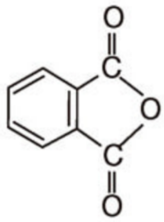
(1) $C_{15}H_{15}NO$

(2)

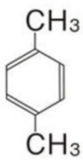
A


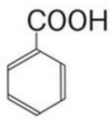
B


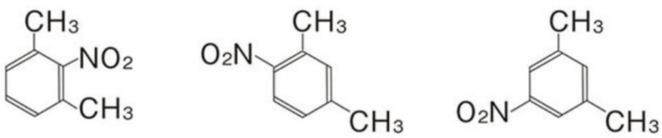
C


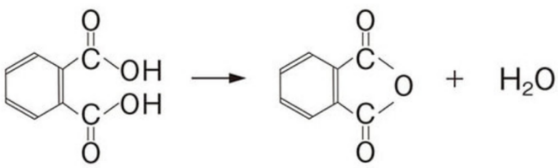
D


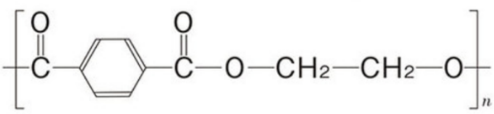
4 – $\boxed{9}$

問 1　化合物 D：


問 2　化合物 E：


問 3


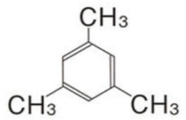
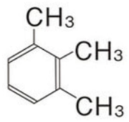
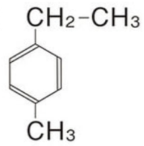
問 4


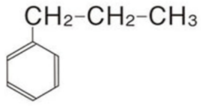
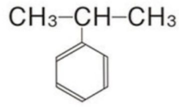
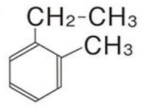
問 5


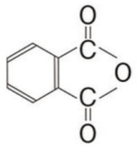
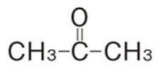
4 – $\boxed{10}$

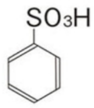
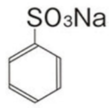
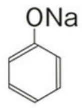
問1（ア） エチレングリコール（1, 2 エタンジオール）　（イ） 酸素

（ウ） ヨウ素

問2 A  B  C 

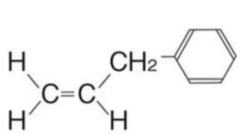
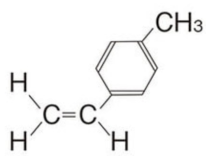
D  E  F 

K  M 

問3 N  O  P 

4 – $\boxed{11}$

問1 2個　**問2** クメン

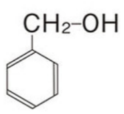
問3A  E 

4－$\boxed{12}$

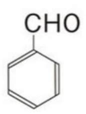
問1 気体…水素　　官能基…ヒドロキシ基

問2 試薬…$FeCl_3$　　色…紫

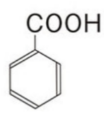
問3 X



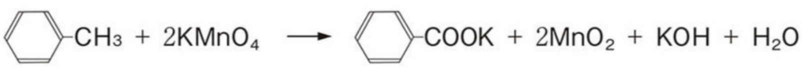
Y



Z

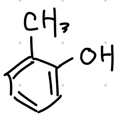


問4

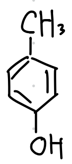


4－$\boxed{13}$

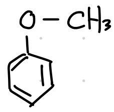
問1　A：



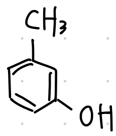
B：



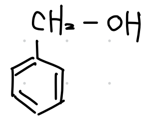
C：



D：



E：



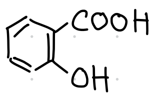
問2　NaOH aq を加えると A,B,D は 中和して 均一な溶液になる（とける）。
（C，E は 2層に分離する）

↑これだけではX。見ためやにおいでわかることをかく。

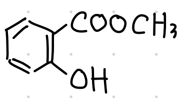
問3　ヒドロキシ基を持たないため，水素結合ができず，分子間力が弱いため。

問4　金属 Na を加えたとき，C は H_2 を発生しない。

問5.　F：サリチル酸

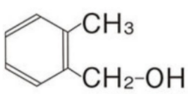


G：サリチル酸メチル

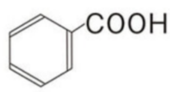


4－$\boxed{14}$

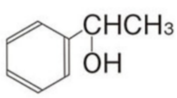
(1)　5　　(2)



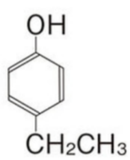
(3)　CHI_3　　(4)



(5)



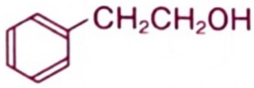
(6)　9　　(7)

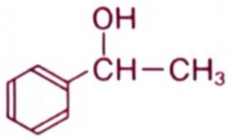


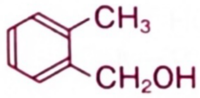
(8)　5　　(9)　1：2：2：2：3

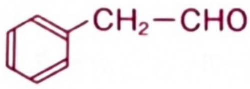
4 − $\boxed{15}$

問1

A


B


C


D


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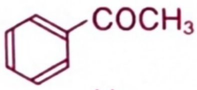
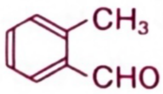


F


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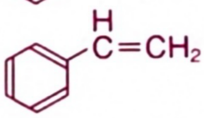
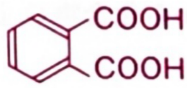


H


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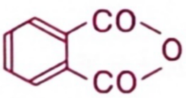
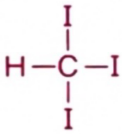
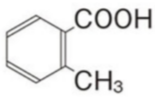


問2　B　　**問3**　ポリスチレン

問4　記号　B　構造式


4 − $\boxed{16}$

問1(1)

A


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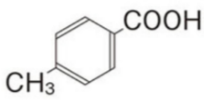
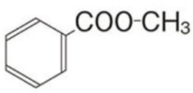


(2)　ナフタレン，*o*−キシレン

問2(1)

C


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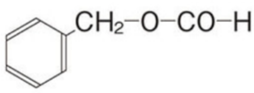

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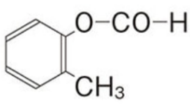
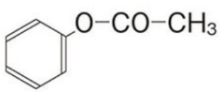


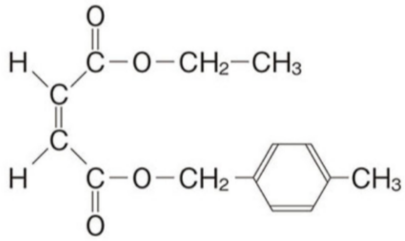
F


(2)　2,4,6−トリブロモフェノール

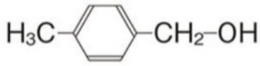
4 − $\boxed{17}$

問1 $C_{14}H_{16}O_4$

問2 A

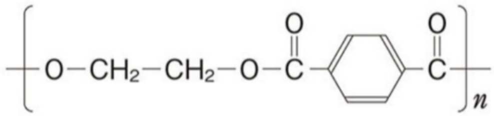


C



D H_3C-CH_2-OH

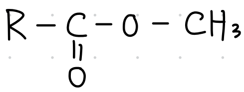
H



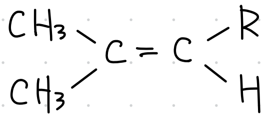
4 − $\boxed{18}$

問1 $C_{12}H_{14}O_2$

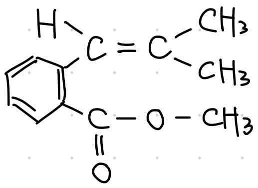
問2 (a)



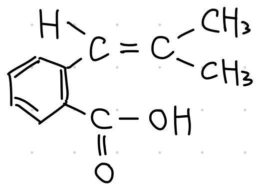
(b)



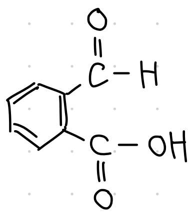
問3 A



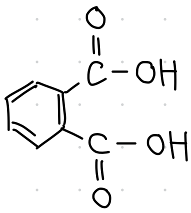
B



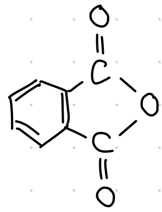
C



D



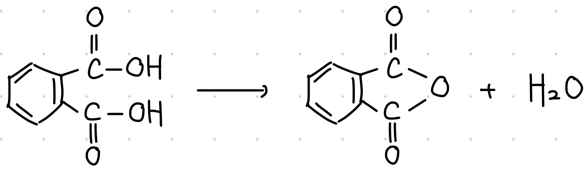
E



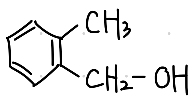
4 − $\boxed{19}$

問1　あ　C_5H_6O　　い　164　　う　$C_{10}H_{12}O_2$

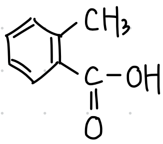
問2



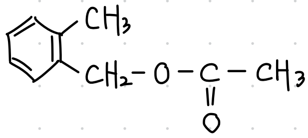
問3　C



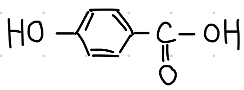
D



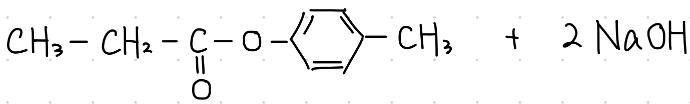
問4.

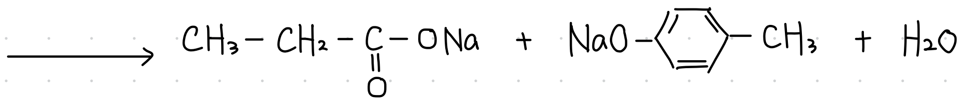


問5



問6



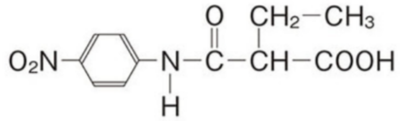


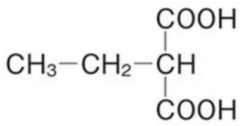
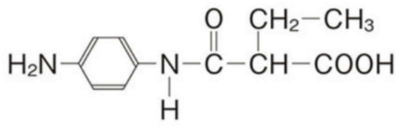
4－$\boxed{20}$

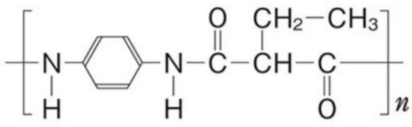
問1　C 原子…52.4 %　　H 原子…4.8 %　　N 原子…11.1 %

問2　$RCOOH + NaHCO_3 \longrightarrow RCOONa + CO_2 + H_2O$

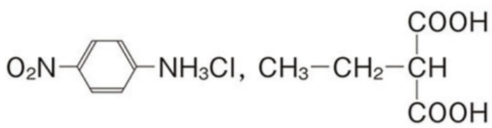
問3

A 　B 

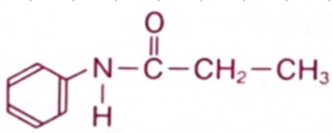
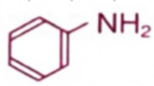
C 　D 

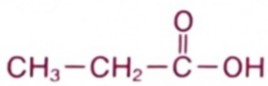
E 

問4　塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

生成物　

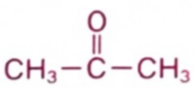
4－$\boxed{21}$

問1　A 　B 

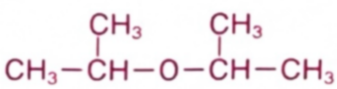
C 　D $CH_3-CH_2-CH_2-OH$

問2　，31 mg

問3　F $CH_3-CH_2-O-CH_3$，エチルメチルエーテル

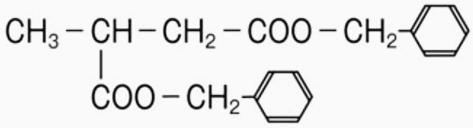
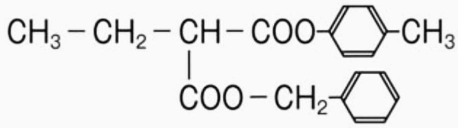
G ，アセトン

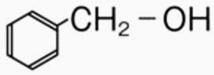
問4　F，E，D

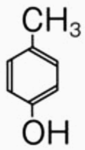
問5　H 　I $CH_2=CH-CH_3$

4 − $\boxed{22}$

問1　$C_{19}H_{20}O_4$

問2　A：　B：

C：　D：$CH_3-\underset{\displaystyle COOH}{\underset{|}{CH}}-CH_2-COOH$

E：　F：$CH_3-CH_2-\underset{\displaystyle COOH}{\underset{|}{CH}}-COOH$　G：